

Automated Supervisory Dispatch Control of a Solar-Grid-Electrolyzer Hydrogen Plant

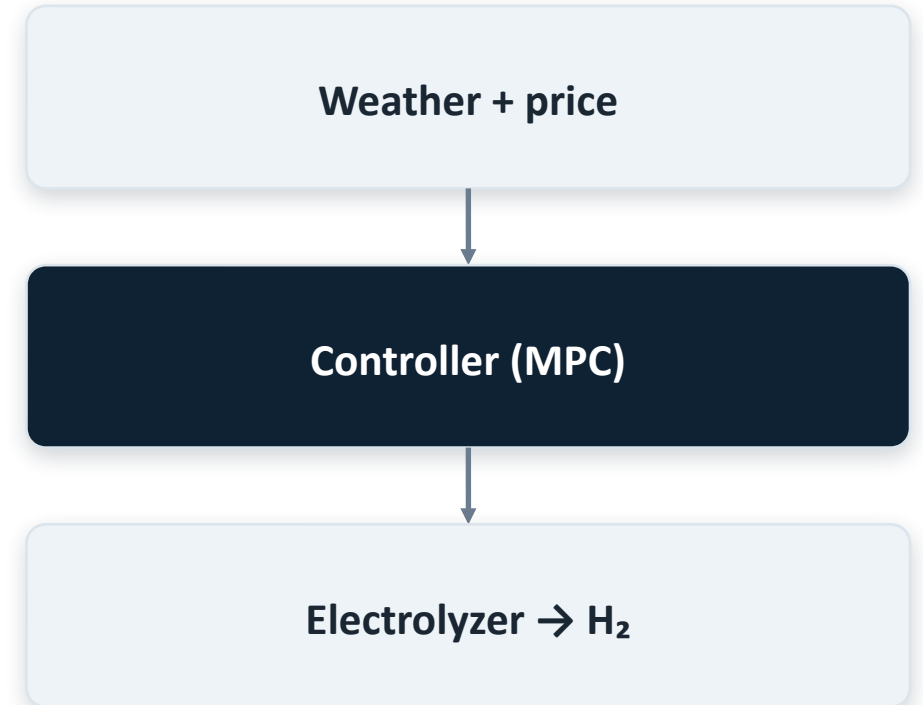
An optimization-based (MPC) controller, validated against an ETAP digital twin and driven by real Ghardaïa weather and the Algerian time-of-use tariff.

Progress update for supervisor

youcefboubidi · June 2026

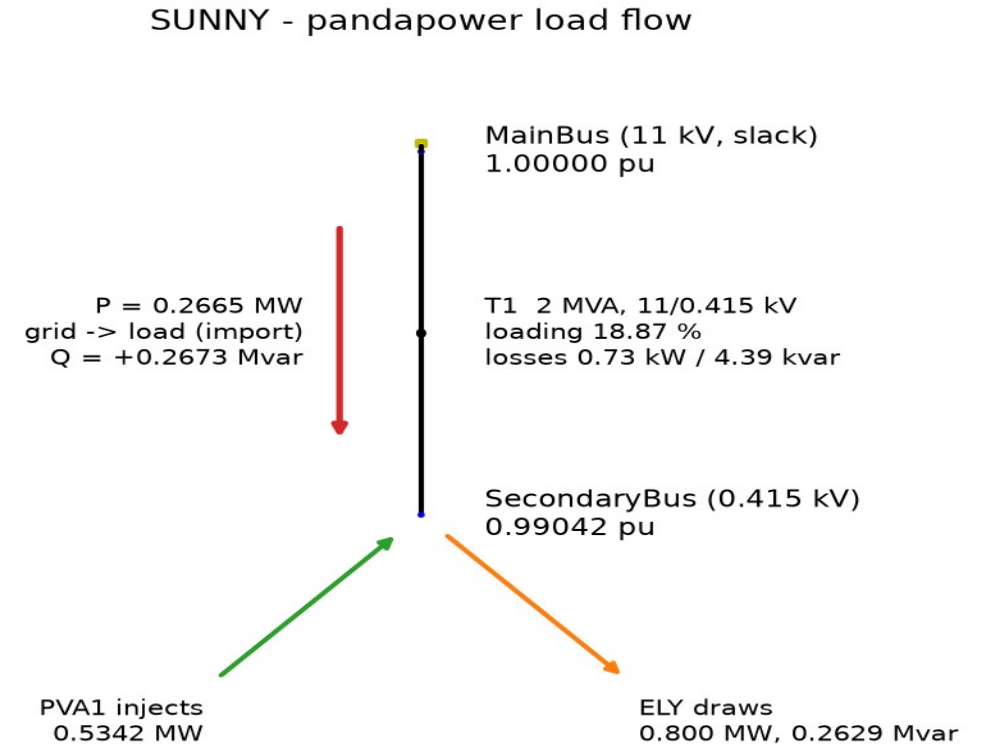
An automatic controller for power dispatch

- Plant: a PEM electrolyzer fed by a solar array and the utility grid, through a 2 MVA / 11–0.415 kV substation.
- Decision, every step: how much power to draw, and from where — under variable irradiance, temperature and electricity price.
- Goal: meet a hydrogen demand at least cost while keeping the network within safe limits (voltage, transformer loading).
- This is a supervisory / optimal-control problem — not a one-off optimization study.



ETAP-validated model, real Ghardaïa inputs

- Electrical network rebuilt in pandapower and validated against the professional ETAP load flow — 24/24 checks pass.
- Solar: real CAMS satellite irradiance + ERA5 temperature → NREL SAM (PySAM), 15-min, full-year 2023.
- Prices: the real Algerian CREG time-of-use tariff (1.21 / 2.16 / 8.11 DA/kWh, night / day / 17–21 h peak).
- Site: Ghardaïa (32.59° N, 3.73° E). Every result rests on measured data, not assumptions.



Receding-horizon (MPC) supervisory controller

State

time, battery charge, cumulative H₂

Control input

electrolyzer setpoint (+ battery power)

Disturbance

PV (weather) — measured, forecast ahead

Constraints

voltage 0.95–1.05 pu, transformer ≤ 100 %, demand

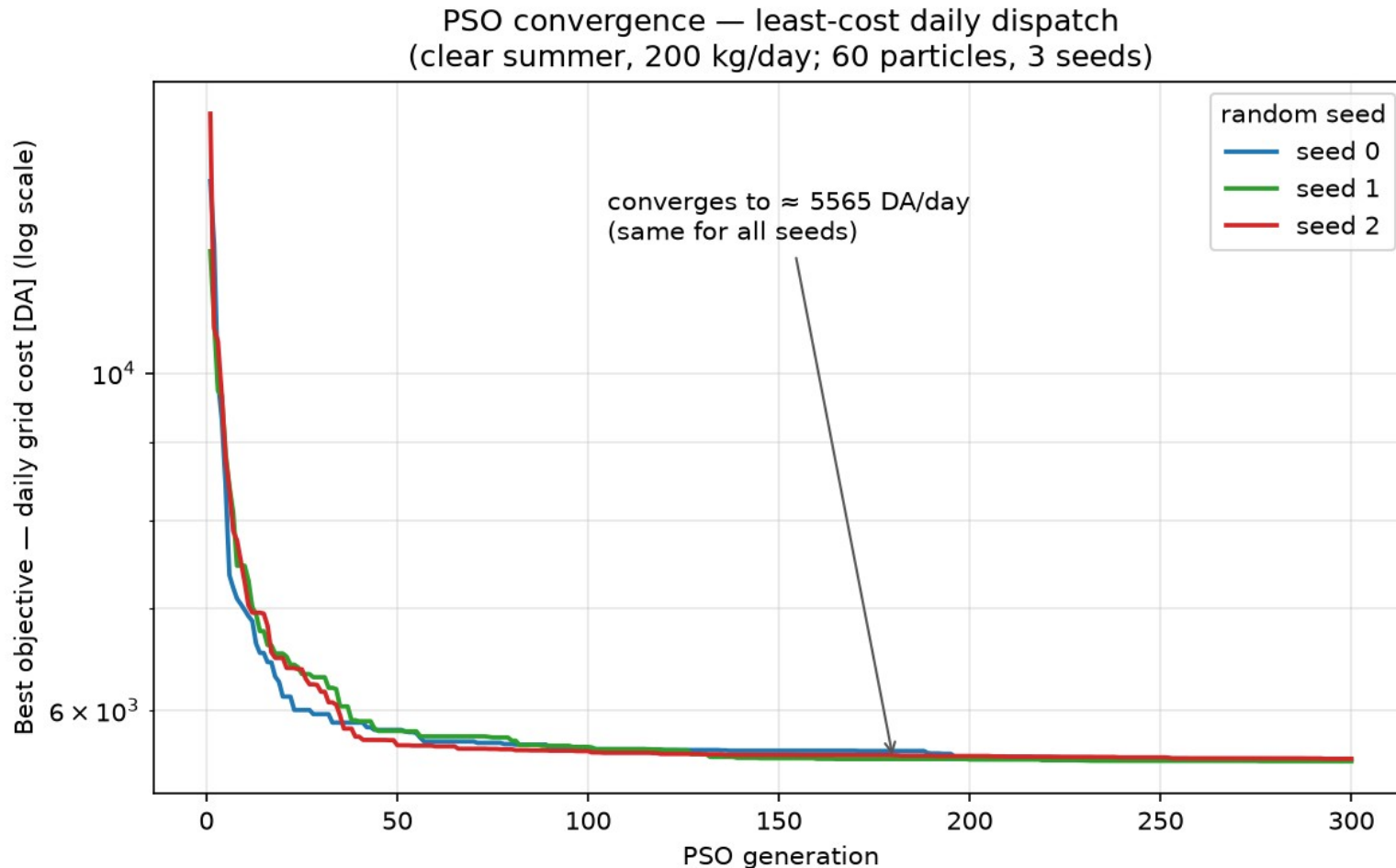
Objective

minimize grid energy cost (time-of-use)

How it works

Each hour the controller re-optimizes the remaining day on a forecast, applies only the next setpoint, then re-plans as the actual PV is realized — closed-loop feedback that rejects the weather disturbance.

How each plan is solved: particle-swarm optimization



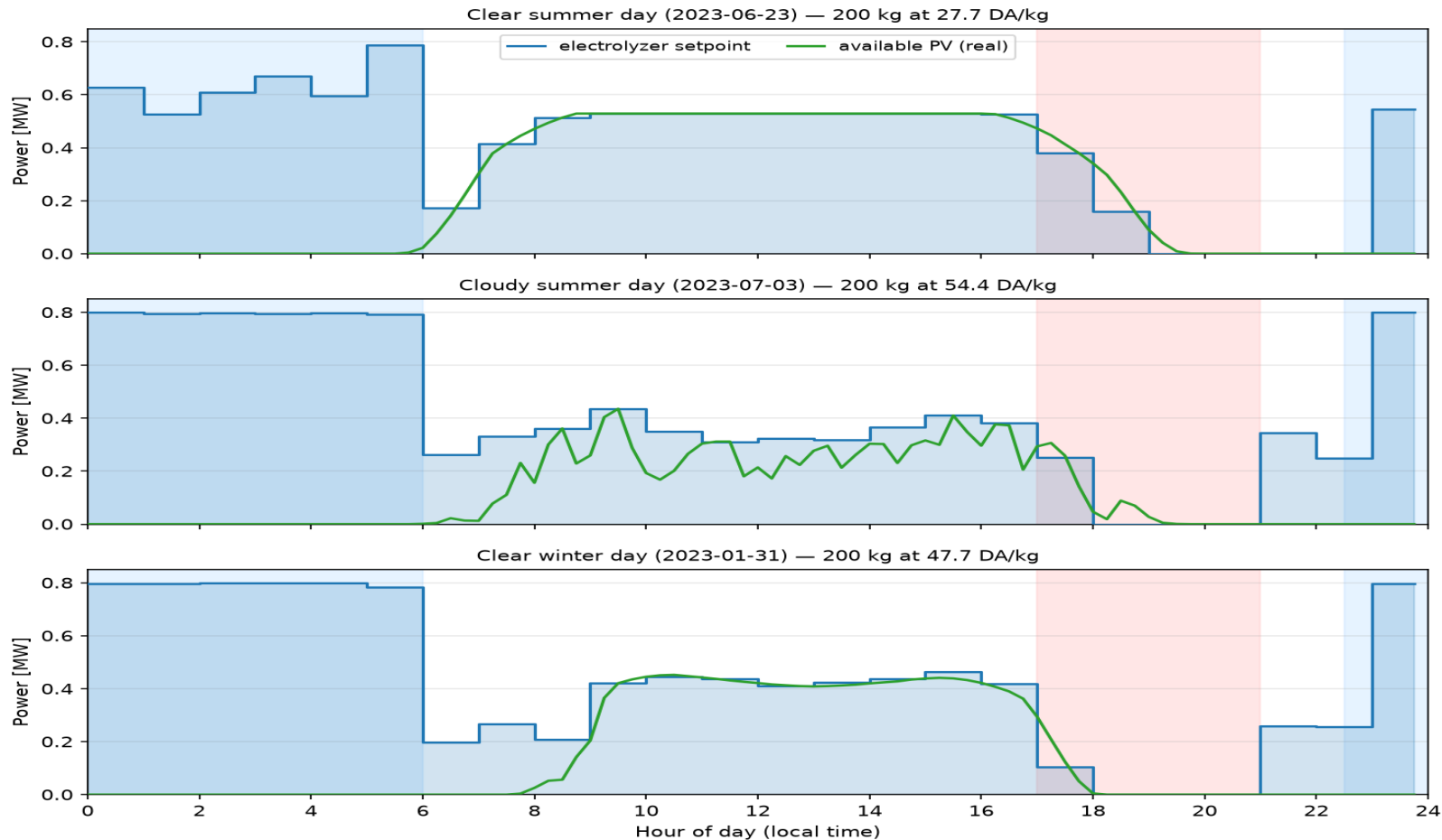
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random seeds reach the same optimum —
reproducible

- Re-planning is a hard search: 24 hourly setpoints, a non-linear electrolyzer and a 3-rate tariff — no closed-form solution.
- PSO = a swarm of candidate day-plans that 'fly' through the search space, each pulled toward its own best and the swarm's best.
- Fitness = grid energy cost + a penalty when the H₂ demand is missed.
- Converges to the least-cost plan in ~ 150 generations.

Rides free solar, dodges the evening peak

Least-cost dispatch to meet 200 kg/day — Ghardaïa (real CAMS+PySAM, 15-min)
blue = off-peak nights, red = 17–21 h peak



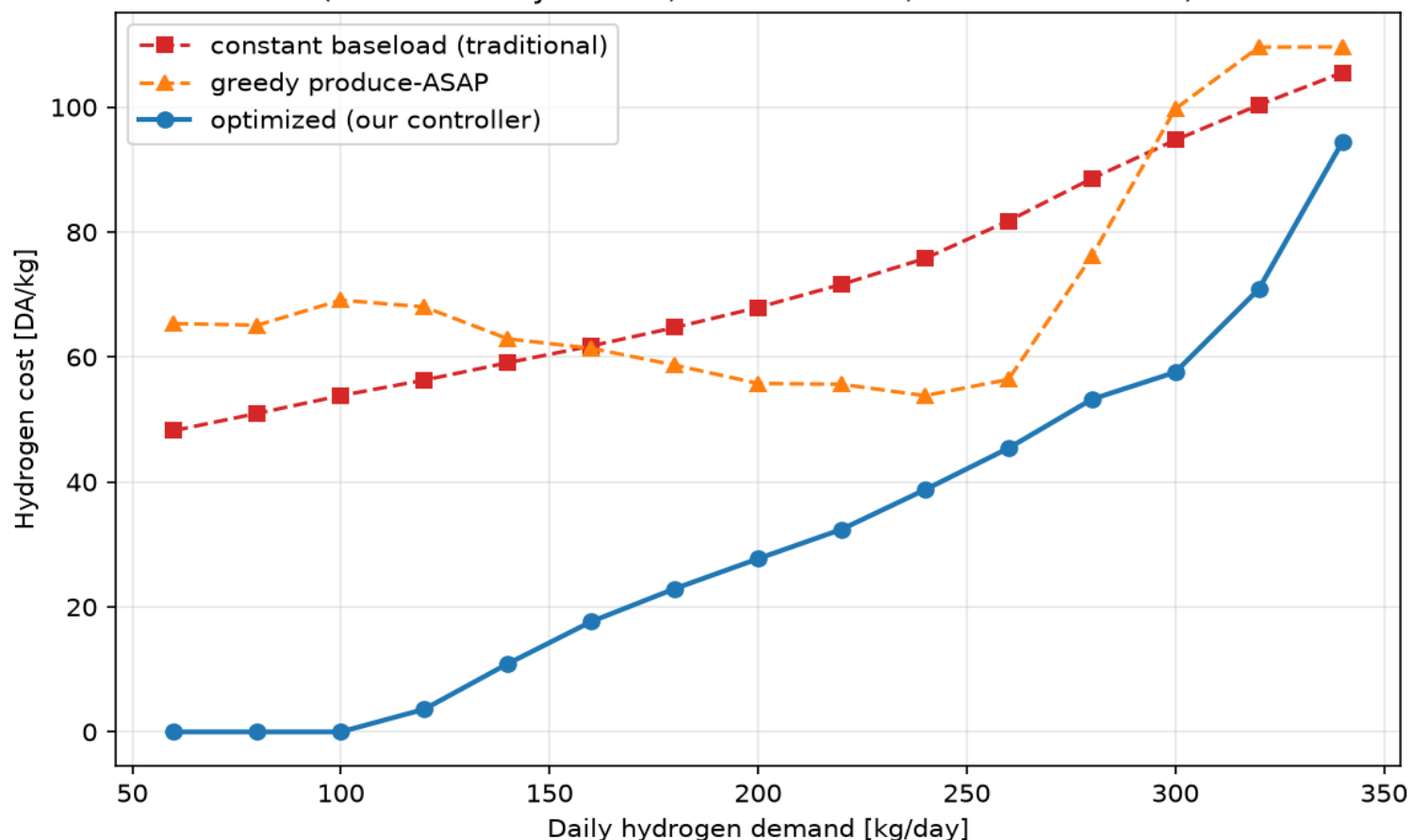
27.8

DA / kg (clear summer, 200 kg/day)

- Full power on cheap night grid (1.21 DA).
- Setpoint tracks the PV mid-day ($\approx$ zero grid).
- Shuts down through the 8.11 DA peak.
- Cloudy day shows real 15-min cloud transients.

50–59 % cheaper than time-blind operation

Smart dispatch vs traditional operation — Ghardaïa clear summer day
(real CAMS+PySAM PV, same demand, same CREG tariff)



–59%

vs constant baseload

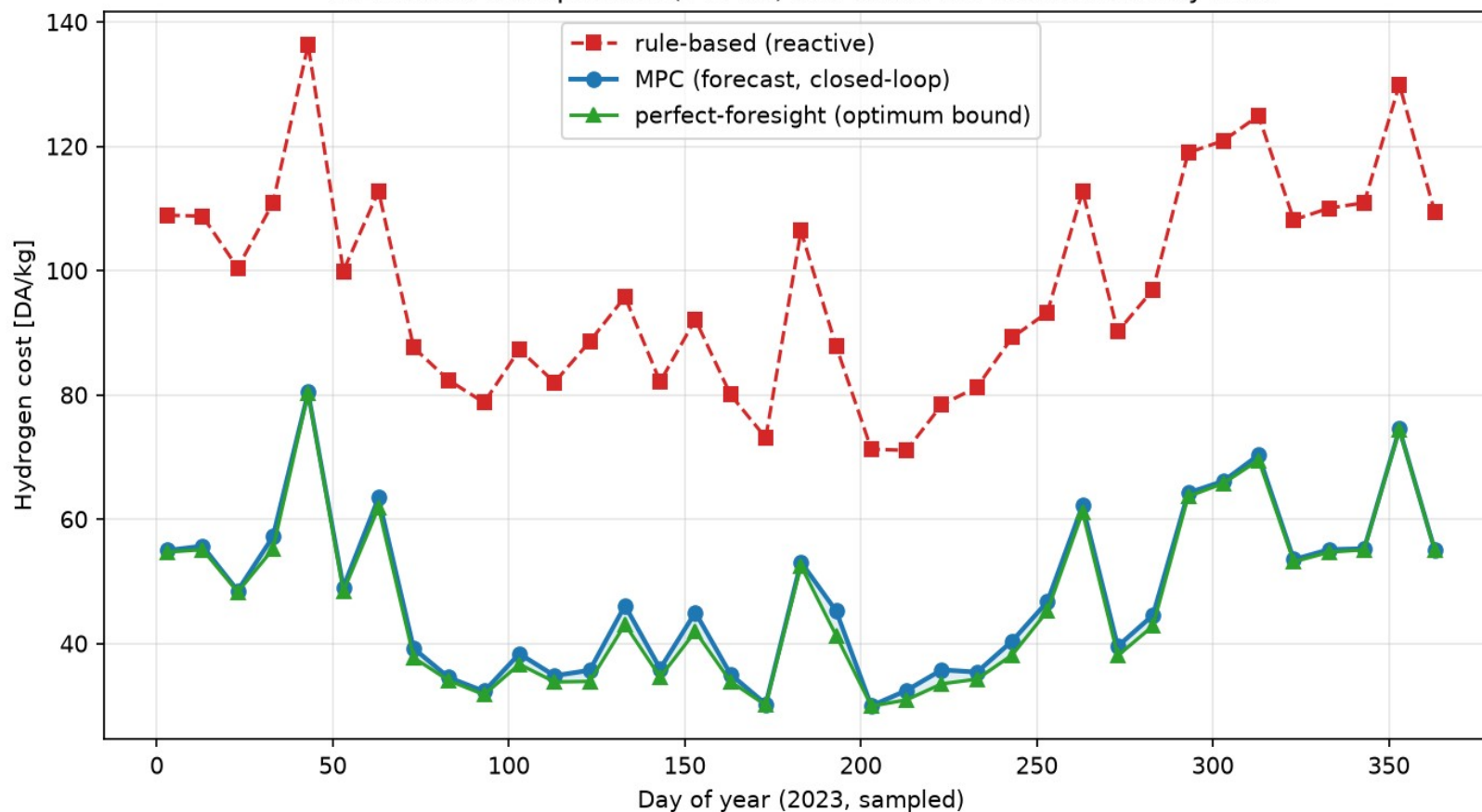
–51%

vs greedy ASAP

- Same hydrogen, same plant, same tariff — only the dispatch timing differs.
- Up to 100 % cheaper at low demand fully covered by solar.
- Naive rules degrade at high demand; the optimizer stays best.

MPC nearly matches the optimum, beats reactive control

Supervisory control strategies — Ghardaïa, 200 kg/day
MPC tracks the optimum (+2.4%) and beats the reactive rule by 51%



+2.4%

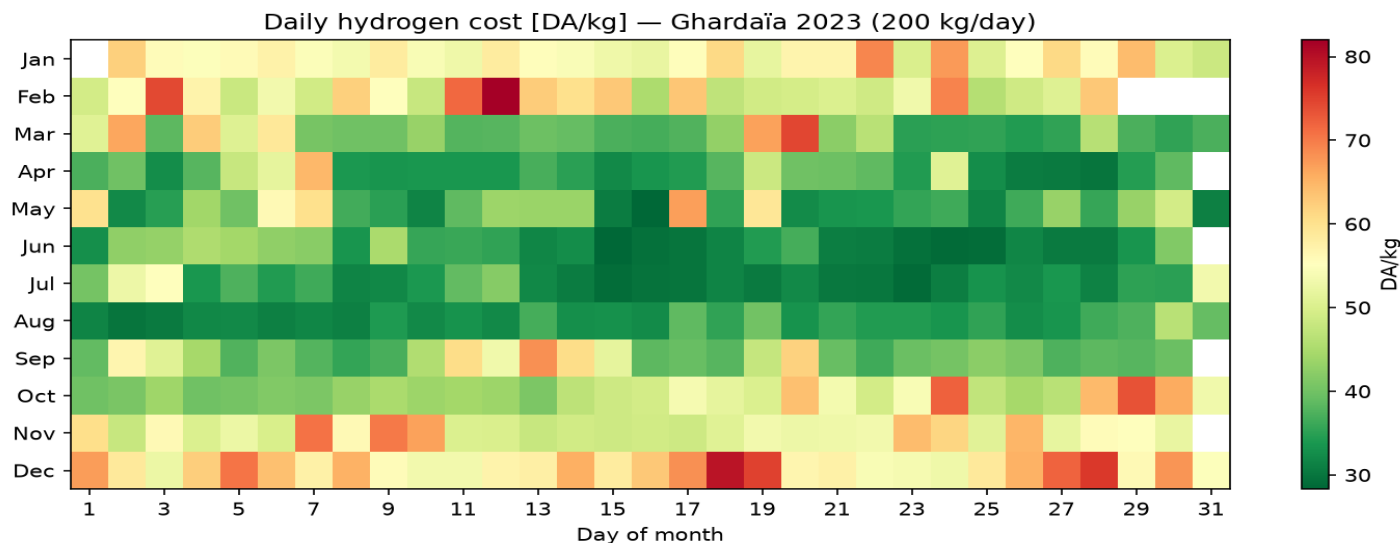
MPC gap to perfect-foresight optimum

-51%

MPC vs reactive rule-based

Closed-loop MPC rejects the PV forecast disturbance and recovers ~98 % of the unreachable optimum — across 37 days of 2023.

Year-round operation and a storage verdict



72.8 t

H₂ / year (2023)

37%

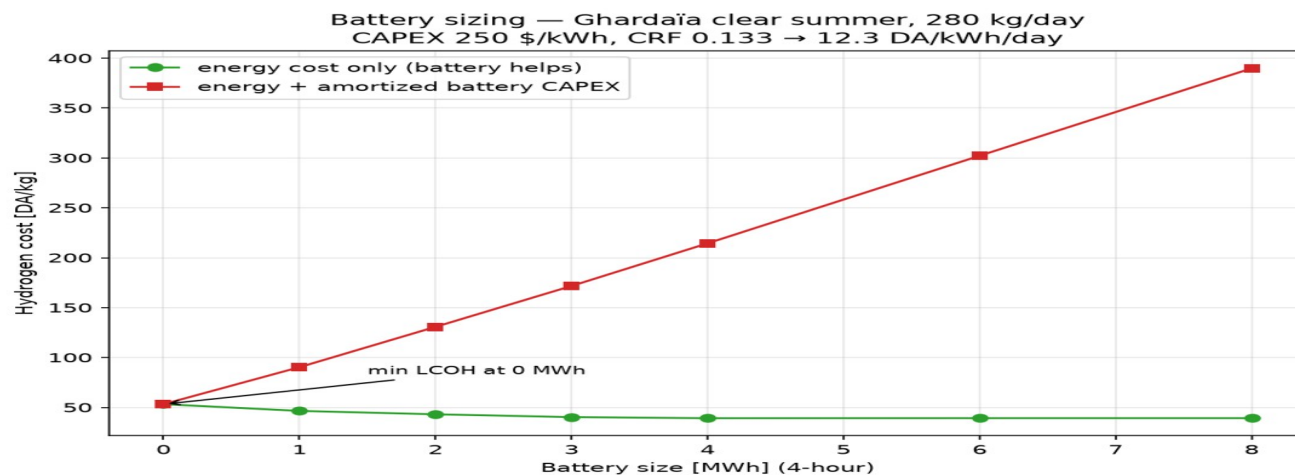
solar-powered

45.8

DA/kg annual avg

0 MWh

optimal battery



- Calendar heatmap: cheap, green summers vs dearer winters.
- Battery cuts energy cost (−26%) but does NOT pay back its CAPEX at this tariff — break-even ≈ \$20/kWh (≈12× below today).
- An honest techno-economic / feasibility finding.

Interactive 3D digital twin



- Three.js 3D scene of the plant with a time slider stepping through the day in 15-minute steps.
- ETAP-style floating labels per device — bus voltage, transformer loading, currents, power — colour-coded green/amber/red against the limits.
- Every value comes from the validated pandapower load flow (no invented numbers). Animated power-flow arrows show import / solar / load.
- Self-contained: opens by double-clicking — works offline, no server.

Where the project stands

DONE

- ETAP-validated network model (24/24 checks).
- Real-data pipeline: CAMS + ERA5 → PySAM, full 2023.
- MPC supervisory controller + strategy comparison.
- Dispatch, full-year and battery results.
- Interactive 3D digital twin (live).

NEXT

- Tighten the control write-up (Ch. 4–5) around the MPC.
- Sensitivity / robustness across more weather years.
- Optional: bind the network (larger PV) so limits activate.
- Polish dashboard + 3D twin for the defense.